

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12411697
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Tadesse; Bersabel et al.

Plugin management system for an interactive system or platform

Abstract

A plugin management system is provided as part of an interactive system, to enable users to search for and execute desired plugins. The plugin management system provides a search user interface to receive inputs from the user, as well as parametric values that are used by the selected plugin. Based on the user interaction with the search user interface, the plugin management system executes identified plugins, using parametric values specified by the user.

Inventors: Tadesse; Bersabel (San Francisco, CA), Sicking; Jonas (San Francisco, CA), Chui; Yi Tang Jackie (San Francisco, CA), Doherty; John (San Francisco, CA)

Applicant: Figma, Inc. (San Francisco, CA)

Family ID: 83283374

Assignee: Figma, Inc. (San Francisco, CA)

Appl. No.: 17/888382

Filed: August 15, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230082639 A1	Mar. 16, 2023

Related U.S. Application Data

us-provisional-application US 63233635 20210816

Publication Classification

Int. Cl.: G06F9/445 (20180101)

U.S. Cl.:

CPC **G06F9/44526** (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5339389	12/1993	Bates	N/A	N/A
7770122	12/2009	Shaik	N/A	N/A
8667480	12/2013	Sigurdsson	717/170	G06F 8/65
9229702	12/2015	Kapulkin	N/A	G06F 8/65
9424545	12/2015	Lee	N/A	N/A
9430229	12/2015	Zijst	N/A	N/A
10606576	12/2019	Tung	N/A	N/A
10620921	12/2019	Brebner	N/A	N/A
11138518	12/2020	Yu	N/A	N/A
2008/0040426	12/2007	Synstelien	N/A	N/A
2010/0192107	12/2009	Takahashi	N/A	N/A
2010/0238494	12/2009	Araki	358/1.15	G06F 3/1204
2010/0306181	12/2009	Little	N/A	N/A
2011/0113350	12/2010	Carlos	N/A	N/A
2011/0268262	12/2010	Jones	N/A	N/A
2012/0159145	12/2011	Cheong	713/100	G06F 8/61
2012/0247967	12/2011	James	N/A	N/A
2013/0111338	12/2012	Huo	N/A	N/A
2014/0310660	12/2013	Rosen	N/A	N/A
2015/0127753	12/2014	Tew	N/A	N/A
2016/0321449	12/2015	Surti	N/A	N/A
2017/0192656	12/2016	Pedrick	N/A	N/A
2017/0262294	12/2016	Yakan	N/A	N/A
2017/0339370	12/2016	Inoue	N/A	N/A
2018/0088915	12/2017	Yang	N/A	G06F 40/186
2018/0181299	12/2017	Ouellette et al.	N/A	N/A
2019/0065440	12/2018	Lu	N/A	G06F 9/44526
2019/0146811	12/2018	Pell	N/A	N/A
2019/0179501	12/2018	Seeley	N/A	N/A
2019/0238602	12/2018	Lo	N/A	N/A
2019/0303880	12/2018	Hashimoto	N/A	N/A
2019/0361580	12/2018	Dowling et al.	N/A	N/A
2020/0042648	12/2019	Rao	N/A	G06F 16/2457
2020/0050431	12/2019	Zilouchian Moghaddam	N/A	G06F 8/38
2020/0296147	12/2019	Eliason	N/A	N/A
2021/0208775	12/2020	Allington	N/A	N/A

2021/0367986	12/2020	Davuluri et al.	N/A	N/A
2022/0108276	12/2021	Stringham	N/A	N/A
2022/0129118	12/2021	Hsu	N/A	N/A
2022/0263675	12/2021	Cupala	N/A	N/A
2022/0334704	12/2021	Lin	N/A	N/A
2022/0334806	12/2021	Lin	N/A	N/A
2022/0342644	12/2021	Fuetsch	N/A	N/A

OTHER PUBLICATIONS

International Search Report and The Written Opinion of The International Searching Authority mailed Nov. 28, 2022, for related PCT/US2022/040461 filed Aug. 16, 2022, 14 pages. cited by applicant

International Search Report and The Written Opinion of The International Searching Authority mailed Dec. 12, 2022, for related PCT/US2022/041418 filed Aug. 24, 2022, 11 pages. cited by applicant

“Zack Qattan, “Collaborative Cursor Chat w/Croquet”, Nov. 30, 2020, youtube.com, <https://www.youtube.com/watch?v=NiqTCROdlgw>, pp. 1-15 (Year: 2020)”. cited by applicant
 Anonymous: “Change cursor into custom text”, Dec. 24, 2018 (Dec. 24, 2018), XP055930588, Retrieved from the Internet: URL: <https://stackoverflow.com/questions/53916492/change-cursor-into-custome-text> [retrieved on Jun. 13, 2022] the whole document, 2 pages. cited by applicant

International Search Report and The Written Opinion of The International Searching Authority mailed Jul. 1, 2022, for related PCT/US2022/025626, filed Apr. 20, 2022, 17 pages. cited by applicant

Wallace Evan: “Multiplayer Editing in Figma”, figma.com/blog, Sep. 28, 2016 (Sep. 28, 2016), XP055930547, Retrieved from the Internet: URL: <https://www.figma.com/blog/multiplayer-editing-in-figma/> [retrieved on Jun. 13, 2022] the whole document, 7 pages. cited by applicant

Anonymous: “HTML editor—Wikipedia”. Apr. 13, 2022 pp. 1-4. cited by applicant

International Search Report and Written Opinion mailed Jun. 27, 2023 in PCT/US2023/015322. cited by applicant

International Search Report dated Aug. 18, 2023 in PCT/US2023/020890. cited by applicant

IPRP dated Jul. 3, 2024 in PCT/US2023/020890. cited by applicant

Written Opinion dated Apr. 10, 2024 in PCT/US2023/020890. cited by applicant

Primary Examiner: Onat; Umut

Attorney, Agent or Firm: Squire Patton Boggs LLP

Background/Summary

RELATED APPLICATION(S) (1) This application claims benefit of priority to Provisional U.S. Patent Application No. 63/233,635, filed Aug. 16, 2021; the aforementioned priority application being hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) Examples described herein relate to graphic design systems, and more specifically, to plugin management for integrated graphic design systems.

BACKGROUND

(2) Software design tools have many forms and applications. In the realm of application user interfaces, for example, software design tools require designers to blend functional aspects of a

program with aesthetics and even legal requirements, resulting in a collection of pages which form the user interface of an application. For a given application, designers often have many objectives and requirements that are difficult to track.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A illustrates an interactive system for a computing device of a user, according to one or more embodiments.
- (2) FIG. 1B illustrates a network computing system to implement an interactive system on a user computing device, according to one or more embodiments.
- (3) FIG. 1C illustrates a network computing system to implement an interactive system for multiple users in a collaborative network platform, according to one or more embodiments.
- (4) FIG. 2 illustrates a plugin management system for use with an interactive system, according to one or more embodiments.
- (5) FIG. 3A illustrates a method for managing plugins in connection with utilizing an interactive system, according to one or more embodiments.
- (6) FIG. 3B illustrates another example method for enabling plugin execution, according to one or more embodiments.
- (7) FIG. 4A through FIG. 4D illustrate an example of a search user interface for selecting and triggering execution of a desired plugin, according to one or more embodiments.
- (8) FIG. 4E and FIG. 4F illustrate another example of a search user interface for selecting and triggering execution of plugins, according to one or more embodiments.
- (9) FIG. 4G illustrates another example of a universal search user interface that enables universal search, including plugin selection and execution, according to one or more embodiments.
- (10) FIG. 4H through FIG. 4J illustrate another example of a search interface to invoke additional functional interfaces for interacting with plugins, according to one or more embodiments.
- (11) FIG. 4K through FIG. 4O illustrate another example of a search interface that utilizes historical information, according to one or more embodiments.
- (12) FIG. 5 illustrates a computer system on which one or more embodiments can be implemented.
- (13) FIG. 6 illustrates a user computing device for use with one or more examples, as described.

DETAILED DESCRIPTION

- (14) Embodiments provide for an interactive system or platform that includes a plugin management system, to enable users to search for and execute desired plugins. In examples, the plugin management system provides a search user interface to receive inputs from the user, as well as parametric values that are used by the selected plugin. Based on the user interaction with the search user interface, the plugin management system executes identified plugins, using parametric values specified by the user.
- (15) In examples, a computing system is configured to implement an interactive system or platform for enabling users to create various types of content, including graphic designs, whiteboards, presentations, web pages and other types of content. Among other advantages, examples as described enable such users to utilize plugins to extend or supplement the functionality of an integrated graphic design system for their particular needs.
- (16) Still further, in some examples, a network computer system is provided to include memory resources store a set of instructions, and one or more processors are operable to communicate the set of instructions to a plurality of user devices. The set of instructions can be communicated to user computing devices, in connection with the user computing devices being operated to render a content on a canvas, where the design can be edited by user input that is indicative of any one of multiple different input actions. The set of instructions can be executed on the computing devices

to cause each of the computing devices to determine one or more input actions to perform based on user input. The instructions may further cause the user computing devices to implement the one or more input actions to modify the content. In such examples, the interactive system includes a plugin management system to enable users to search for and execute plugins that extend or supplement the functionality provided by the plugin management system.

(17) One or more embodiments described herein provide that methods, techniques, and actions performed by a computing device are performed programmatically, or as a computer-implemented method. Programmatically, as used herein, means through the use of code or computer-executable instructions. These instructions can be stored in one or more memory resources of the computing device. A programmatically performed step may or may not be automatic.

(18) One or more embodiments described herein can be implemented using programmatic modules, engines, or components. A programmatic module, engine, or component can include a program, a sub-routine, a portion of a program, or a software component or a hardware component capable of performing one or more stated tasks or functions. As used herein, a module or component can exist on a hardware component independently of other modules or components. Alternatively, a module or component can be a shared element or process of other modules, programs or machines.

(19) Some embodiments described herein can generally require the use of computing devices, including processing and memory resources. For example, one or more embodiments described herein may be implemented, in whole or in part, on computing devices such as servers, desktop computers, cellular or smartphones, tablets, wearable electronic devices, laptop computers, printers, digital picture frames, network equipment (e.g., routers) and tablet devices. Memory, processing, and network resources may all be used in connection with the establishment, use, or performance of any embodiment described herein (including with the performance of any method or with the implementation of any system).

(20) Furthermore, one or more embodiments described herein may be implemented through the use of instructions that are executable by one or more processors. These instructions may be carried on a computer-readable medium. Machines shown or described with figures below provide examples of processing resources and computer-readable mediums on which instructions for implementing embodiments of the invention can be carried and/or executed. In particular, the numerous machines shown with embodiments of the invention include processor(s) and various forms of memory for holding data and instructions. Examples of computer-readable mediums include permanent memory storage devices, such as hard drives on personal computers or servers. Other examples of computer storage mediums include portable storage units, such as CD or DVD units, flash memory (such as carried on smartphones, multifunctional devices or tablets), and magnetic memory.

Computers, terminals, network enabled devices (e.g., mobile devices, such as cell phones) are all examples of machines and devices that utilize processors, memory, and instructions stored on computer-readable mediums. Additionally, embodiments may be implemented in the form of computer-programs, or a computer usable carrier medium capable of carrying such a program.

(21) System Description

(22) FIG. 1A illustrates an interactive system for a computing device of a user, according to one or more examples. An interactive system **100** can be implemented in any one of multiple different computing environments. For example, in some variations, the system **100** can be implemented as a client-side application that executes on the user computing device **10** to provide functionality as described with various examples. In other examples, such as described below, the system **100** can be implemented through use of a web-based application **80**. As an addition or alternative, the system **100** can be implemented as a distributed computing environment, such that processes described with various examples execute on a network computer (e.g., server) and/or on the user device **10**.

(23) According to examples, interactive system **100** is implemented on a user computing device **10** to enable a corresponding user to generate content such as interactive designs and whiteboards. The

system **100** can include processes that execute as or through a web-based application **80** that is installed on the computing device **10**. As described by various examples, web-based application **80** can execute scripts, code and/or other logic (the “programmable components”) to implement functionality of the interactive system **100**. Additionally, in some variations, the system **100** can be implemented as part of a network service, where web-based application **80** communicates with one or more remote computers (e.g., server used for a network service) to executes processes of the system **100**.

(24) In some examples, web-based application **80** retrieves some or all of the programmatic resources for implementing the system **100** from a network site. As an addition or alternative, web-based application **80** can retrieve some or all of the programmatic resources from a local source (e.g., local memory residing with the computing device **10**). The web-based application **80** may also access various types of data sets in providing functionality or services for the interactive system **100**. The data sets can correspond to files and libraries, which can be stored remotely (e.g., on a server, in association with an account), locally or distributed between local and network resources.

(25) In examples, the web-based application **80** can correspond to a commercially available browser, such as GOOGLE CHROME (developed by GOOGLE, INC.), SAFARI (developed by APPLE, INC.), and INTERNET EXPLORER (developed by the MICROSOFT CORPORATION). In such examples, the processes of the interactive system **100** can be implemented as scripts and/or other embedded code which web-based application **80** downloads from a network site. For example, the web-based application **80** can execute code that is embedded within a webpage to implement processes of the system **100**. The web-based application **80** can also execute the scripts to retrieve other scripts and programmatic resources (e.g., libraries) from the network site and/or other local or remote locations. By way of example, the web-based application **80** may execute JAVASCRIPT embedded in an HTML resource (e.g., web-page structured in accordance with HTML 5.0 or other versions, as provided under standards published by W3C or WHATWG consortiums). In some examples, the rendering engine **120** and/or other components may utilize graphics processing unit (GPU) accelerated logic, such as provided through WebGL (Web Graphics Library) programs which execute Graphics Library Shader Language (GLSL) programs that execute on GPUs.

(26) According to examples, the user of computing device **10** operates web-based application **80** to access a network site, where programmatic resources are retrieved and executed to implement the interactive system **100**. In some examples, the user may initiate a session to implement the interactive system **100** for purpose of creating and/or editing a graphic design, whiteboard, presentation, a webpage or other type of content. In examples, the system **100** includes a program interface **102**, an input interface **118**, and a rendering engine **120**. The program interface **102** can include one or more processes which execute to access and retrieve programmatic resources from local and/or remote sources.

(27) In an implementation, the program interface **102** can generate, for example, a canvas **122**, using programmatic resources which are associated with web-based application **80** (e.g., HTML 5.0 canvas). As an addition or variation, the program interface **102** can trigger or otherwise cause the canvas **122** to be generated using programmatic resources and data sets (e.g., canvas parameters) which are retrieved from local (e.g., memory) or remote sources (e.g., from network service).

(28) The program interface **102** may also retrieve programmatic resources that include an application framework for use with canvas **122**. The application framework can include data sets which define or configure, for example, a set of interactive tools that integrate with the canvas **122** and which comprise the input interface **118**, to enable the user to provide input for creating and/or editing a given content (e.g., a graphic design, a whiteboard, a presentation, a webpage, etc.).

(29) According to some examples, the input interface **118** can be implemented as a functional layer that is integrated with the canvas **122** to detect and interpret user input. The input interface **118** can,

for example, use a reference of the canvas **122** to identify a screen location of a user input (e.g., 'click'). Additionally, the input interface **118** can interpret an input action of the user based on the location of the detected input (e.g., whether the position of the input indicates selection of a tool, an object rendered on the canvas, or region of the canvas), the frequency of the detected input in a given time period (e.g., double-click), and/or the start and end position of an input or series of inputs (e.g., start and end position of a click and drag), as well as various other input types which the user can specify (e.g., right-click, screen-tap, etc.) through one or more input devices. In this manner, the input interface **118** can interpret, for example, a series of inputs as a design tool selection (e.g., shape selection based on location of input), as well as inputs to define attributes (e.g., dimensions) of a selected shape.

(30) Additionally, the program interface **102** can be used to retrieve, from local or remote sources, programmatic resources and data sets which include files **101** which comprise an active workspace for the user. The retrieved data sets can include, for example, one or more pages that include content elements which collectively form a given content. By way of example, the content can correspond to a design interface, whiteboard, webpage, or other content medium. Each file **101** can include one or multiple data structure representations **111** which collectively define the design interface. The files **101** may also include additional data sets which are associated with the active workspace. For example, as described with some examples, the individual pages of the active workspace may be associated with a set of constraints **145**. As an additional example, the program interface **102** can retrieve (e.g., from network service **152** (see FIG. **1B**), from local memory, etc.) one or more types of profile information **109**, such as user profile information which can identify past activities of the user of the computing device **10** when utilizing the interactive system **100**. The profile information **109** can identify, for example, input types (or actions) of the user with respect to the page(s) of the active workspace, or more generally, input actions of the user in a prior time interval. In some variations, the profile information **109** can also identify historical or contextual information about individual design interfaces, as represented by corresponding data structure representations **111**.

(31) In examples, the rendering engine **120** uses the data structure representations **111** to render a corresponding content **125** on the canvas **122**, wherein the content **125** reflects elements or components and their respective attributes, as may be provided with the individual pages of the files **101**. The user can edit the content **125** using the input interface **118**. Alternatively, the rendering engine **120** can generate a blank page for the canvas **122**, and the user can use the input interface **118** to generate the content **125**. By way of example, the content **125** can include graphic elements such as a background and/or a set of objects (e.g., shapes, text, images, programmatic elements), as well as attributes of the individual graphic elements. Each attribute of a graphic element can include an attribute type and an attribute value. For an object, the types of attributes include, shape, dimension (or size), layer, type, color, line thickness, text size, text color, font, and/or other visual characteristics. Depending on implementation, the attributes reflect properties of two- or three-dimensional designs. In this way, attribute values of individual objects can define, for example, visual characteristics of size, color, positioning, layering, and content, for elements that are rendered as part of the content **125**.

(32) In examples, individual design elements may also be defined in accordance with a desired run-time behavior. By way of example, some objects can be defined to have run-time behaviors that are either static or dynamic. The attributes of dynamic objects may change in response to predefined run-time events generated by the underlying application that is to incorporate the content **125**. Additionally, some objects may be associated with logic that defines the object as being a trigger for rendering or changing other objects, such as through implementation of a sequence or workflow. Still further, other objects may be associated with logic that provides the design elements to be conditional as to when they are rendered and/or their respective configuration or appearance when rendered. Still further, objects may also be defined to be interactive, where one or

more attributes of the object may change based on user-input during the run-time of the application.

(33) As described with examples, the interactive system **100** enables the user of plugins by users. A plugin can be selected and executed to perform a specific set of operations, and execution of the plugin can alter the content **125** on the canvas **122**. For example, a plugin library can be stored on the user computing device **10** and/or stored on a network site which the interactive system **100**. Further, in examples, plugins can be used to perform a task that is difficult or time-consuming. For example, in implementations where the system **100** enables creation of interactive graphic designs, plugins can be executed to create specific types of content graphic content elements (e.g., generate iconic representation of person, create interactive table, etc.). Still further, a plugin can be configured to perform a task of altering attributes of content elements. For example, a plugin can execute to implement a task that automatically replaces the occurrence of an attribute (e.g., fill color, line color, etc.) with another attribute. Still further, plugins can implement other types of tasks, such as exporting content elements or creating data sets (e.g., programmatic code) for specified content elements. Such examples illustrate the various ways plugins can be incorporated and used with an interactive system **100**, such as described by various examples.

(34) Network Computing System

(35) FIG. **1B** illustrates a network computing system to implement an interactive system on a user computing device, according to one or more examples. A network computing system such as described with an example of FIG. **1B** can be implemented using, for example, one or more servers which communicate with user computing devices over one or more networks.

(36) In an example of FIG. **1B**, the network computing system **150** perform operations to enable the interactive system **100** to be implemented on the user computing device **10**. In variations, the network computing system **150** provides a network service **152** to support the use of the interactive system **100** by user computing devices that utilize browsers or other web-based applications. The network computing system **150** can include a site manager **158** to manage a website where a set of web-resources **155** (e.g., web page) are made available for site visitors. The web-resources **155** can include instructions, such as scripts or other logic (“system instructions **157**”), which are executable by browsers or web components of user computing devices.

(37) In some variations, once the computing device **10** accesses and downloads the web-resources **155**, web-based application **80** executes system instructions **157** to implement functionality such as described with some examples of FIG. **1A**. For example, the system instructions **157** can be executed by web-based application **80** to initiate the program interface **102** on the user computing device **10**. The initiation of the program interface **102** may coincide with the establishment of, for example, a web-socket connection between the program interface **102** and a service component **160** of the network computing system **150**.

(38) In some examples, the web-resources **155** includes logic which web-based application **80** executes to initiate one or more processes of the program interface **102**, causing the interactive system **100** to retrieve additional programmatic resources and data sets for implementing functionality as described by examples. The web resources **155** can, for example, embed logic (e.g., JAVASCRIPT code), including GPU accelerated logic, in an HTML page for download by computing devices of users. The program interface **102** can be triggered to retrieve additional programmatic resources and data sets from, for example, the network service **152**, and/or from local resources of the computing device **10**, in order to implement the interactive system **100**. For example, some of the components of the interactive system **100** can be implemented through web-pages that can be downloaded onto the computing device **10** after authentication is performed, and/or once the user performs additional actions (e.g., download one or more pages of the workspace associated with the account identifier). Accordingly, in examples as described, the network computing system **150** can communicate the system instructions **157** to the computing device **10** through a combination of network communications, including through downloading activity of web-based application **80**, where the system instructions **157** are received and executed

by web-based application **80**.

(39) The computing device **10** can use web-based application **80** to access a website of the network service **152** to download the webpage or web resource. Upon accessing the website, web-based application **80** can automatically (e.g., through saved credentials) or through manual input, communicate an account identifier to the service component **160**. In some examples, web-based application **80** can also communicate one or more additional identifiers that correlate to a user identifier.

(40) Additionally, in some examples, the service component **160** can use the user or account identifier of the user identifier to retrieve profile information **109** from a user profile store **166**. As an addition or variation, profile information **109** for the user can be determined and stored locally on the user's computing device **10**.

(41) The service component **160** can also retrieve the files of an active workspace ("active workspace files **163**") that are linked to the user account or identifier from a file store **164**. The profile store **166** can also identify the workspace that is identified with the account and/or user, and the file store **164** can store the data sets that comprise the workspace. The data sets stored with the file store **164** can include, for example, the pages of a workspace, data sets that identify constraints for an active set of workspace files, and one or more data structure representations **161** for the design under edit which is renderable from the respective active workspace files.

(42) Additionally, in examples, the service component **160** provides a representation **159** of the workspace associated with the user to the web-based application **80**, where the representation identifies, for examples, individual files associated with the user and/or user account. The workspace representation **159** can also identify a set of files, where each file includes one or multiple pages, and each page including objects that are part of a design interface.

(43) On the user device **10**, the user can view the workspace representation through web-based application **80**, and the user can elect to open a file of the workspace through web-based application **80**. In examples, upon the user electing to open one of the active workspace files **163**, web-based application **80** initiates the canvas **122**. For example, the interactive system **100** can initiate an HTML 5.0 canvas as a component of web-based application **80**, and the rendering engine **120** can access one or more data structures representations **111** to render or update the corresponding content **125** on the canvas **122**.

(44) With further reference to FIG. 1B, the network computing system **150** enables the user computing device **10** to implement a plugin management system **200**. For example, the network computing system **150** can provide the computing device **10** with logic to cause the computing device **10** to implement the plugin management system **200**. Further, in some examples, the network computing system **150** can store a library or collection of plugins that are made available to individual users through a search user interface, such as described with other examples. In this way, the user of computing device **10** can search for and execute desired plugins to extend or supplement the functionality of the interactive system **100**.

(45) Collaborative Network Platform

(46) FIG. 1C illustrates a network computing system to implement an interactive system for multiple users in a collaborative network platform, according to one or more examples. In an example of FIG. 1C, a collaborative network platform is implemented by the network computing system **150**, which communicates with multiple user computing devices **10**, **12** over one or more networks (e.g., World Wide Web) to implement the interactive system **100** on user computing devices **10**, **12**. While FIG. 1C illustrates an example in which two users utilize the collaborative network platform, examples as described allow for the network computing system **150** to enable collaboration on design interfaces amongst a larger group of users.

(47) With respect to FIG. 1C, the user computing devices **10**, **12** can be assumed as being operated by users that are associated with a common account, with each user computing device **10**, **12** implementing an instance of the interactive system **100** to access the same workspace during

respective sessions that overlap with one another. Accordingly, each of the user computing devices **10, 12** may access the same set of active workspace files **163** at the same time, with the respective program interface **102** of the interactive system **100** on each user computing device **10, 12** operating to establish a corresponding communication channel (e.g., web socket connection) with the service component **160**.

(48) In examples, the service component **160** can communicate a copy of the active workspace files **163** to each user computing device **10, 12**, such that the computing devices **10, 12** render the content **125** of the active workspace files **163** at the same time. Additionally, each of the computing devices **10, 12** can maintain a local data structure representation **111** of the respective content **125**, as determined from the active workspace files **163**. The service component **160** can also maintain a network-side data structure representation **161** obtained from the files of the active workspace **163**, and coinciding with the local data structure representations **111** on each of the computing devices **10, 12**.

(49) The network computing system **150** can continuously synchronize the active workspace files **163** on each of the user computing devices. In particular, changes made by each user to the content **125** on their respective computing device **10, 12** can be immediately reflected on the content **125** rendered on the other user computing device **10, 12**. By way of example, the user of computing device **10** can make a change to the respective content **125**, and the respective rendering engine **120** can implement an update that is reflected in the local copy of the data structure representation **111**. From the computing device **10**, the program interface **102** of the interactive system **100** can stream change data **121**, reflecting the change of the user input, to the service component **160**. The service component **160** processes the change data **121** of the user computing device. The service component **160** can use the change data **121** to make a corresponding change to the network-side data structure representation **161**. The service component **160** can also stream remotely-generated change data **171** (which in the example provided, corresponds or reflects change data **121** received from the user device **10**) to the computing device **12**, to cause the corresponding instance of the interactive system **100** to update the content **125** as rendered on that device. The computing device **12** may also use the remotely generated change data **171** to update with the local data structure representation **111** of that computing device **12**. The program interface **102** of the computing device **12** can receive the update from the network computing system **150**, and the rendering engine **120** can update the content **125** and the respective local copy of **111** of the computing device **12**.

(50) The reverse process can also be implemented to update the data structure representations **161** of the network computing system **150** using change data **121** communicated from the second computing device **12** (e.g., corresponding to the user of the second computing device updating the content **125** as rendered on the second computing device **12**). In turn, the network computing system **150** can stream remotely generated change data **171** (which in the example provided, corresponds or reflects change data **121** received from the user device **12**) to update the local data structure representation **111** of the content **125** on the first computing device **10**. In this way, the content **125** of the first computing device **10** can be updated as a response to the user of the second computing device **12** providing user input to change the content **125**.

(51) To facilitate the synchronization of the data structure representations **111, 111** on the computing devices **10, 12**, the network computing system **150** may implement a stream connector to merge the data streams which are exchanged between the first computing device **10** and the network computing system **150**, and between the second computing device **12** and the network computing system **150**. In some implementations, the stream connector can be implemented to enable each computing device **10, 12** to make changes to the network-side data representation **161**, without added data replication that may otherwise be required to process the streams from each device separately.

(52) Additionally, over time, one or both of the computing devices **10, 12** may become out-of-sync with the server-side data representation **161**. In such cases, the respective computing device **10, 12**

can redownload the active workspace files **163**, to restart its maintenance of the data structure representation of the content **125** that is rendered and edited on that device.

(53) With further reference to FIG. **1C**, the network computing system **150** enables each computing device **10**, **12** to implement a plugin management system **200**. For example, the network computing system **150** can communicate instructions to cause each computing device **10**, **12** to implement the plugin management system **200** as part of the interactive system **100**. Further, in some examples, the network computing system **150** can store a library or collection of plugins that are made available to users of computing devices **10**, **12** through a search user interface, such as described with other examples. In this way, the user can search for and execute desired plugins to extend or supplement the functionality of the interactive system **100**.

(54) Plugin Management System

(55) FIG. **2** illustrates a plugin management system for use with examples as described with FIG. **1A** through FIG. **1C**. In some examples, a plugin management system **200** can be provided as part of the interactive system **100**. In variations, the plugin management system **200** can be provided by, for example, the network computing system **150**, in connection with users utilizing interactive system **100** on their respective computing devices **10**, **12**. As described with various examples, the plugin management system **200** enables a search and execution process for plugins, in connection with users creating and editing designs using the interactive system **100**. Through the search and execution process, the user can execute a desired plugin in an intended manner, with the plugin's execution affecting a design, canvas or other aspect of the interactive system **100**.

(56) According to examples, the system **200** implements the search and execution process to seamlessly transition the user from providing search input (search terms for searching/identifying a desired plugin) to providing plugin input (input for identifying desired parametric values to successfully execute the identified plugin). For example, the system **200** can enable the user to provide a successive series of inputs through a user interface of the system **200** (e.g., using a keyboard) to cause a desired plugin from a plugin library **250** to execute using desired parametric values of the user. Among other advantages, the system **200** enables a marketplace of third-party plugins to be provided to a user, using an interface that enables the user to search and execute desired plugins with a minimal number of inputs that can be successively provided through a common interface or input mechanism (e.g., keyboard). In this way, the system **200** enables the plugin search and execution process to be performed as a continuous or repeated user interaction. The result is a highly efficient interaction for the user, where the user can execute a desired plugin in a manner intended using a keyboard and/or common user interface. Moreover, the system **200** can implement the search and execution process in a manner that ensures input data for plugin execution is appropriately formatted or structured for a particular plugin that is identified through a search process, with no a priori knowledge of the plugins operations being required from the user.

(57) In context of embodiments described, a plugin can correspond to a program that can execute on an end user device to provide additional or enhanced functionality to the interactive system **100**. An end user can, for example, execute a plugin in connection with utilizing the interactive system **100** and creating or updating a design. Among other advantages, examples such as described below enable a user to search for and trigger execution of plugins using alphanumeric entry, such as provided through use of a keyboard. Thus, for example, a user can continuously or continuously use a keyboard to search for and trigger execution of individual plugins.

(58) According to examples, plugin management system **200** includes a search user interface **210**, a search component **220**, a plugin interface component **230**, a search index **240**, a plugin library **250** and a developer interface **260**. Depending on implementation, the plugin management system **200** can be implemented on a user computing device **10**, **12**, on a network computer system **150**, and/or distributed between a user computing device **10**, **12** and the network computing system **150**.

(59) Search Interface to Receive Search and Plugin Input

(60) In examples, the search user interface **210** can be rendered through implementation of the

interactive system **100**. For example, the search user interface **210** can be provided as part of a design panel, or as a feature the user can access and use with a canvas. According to examples, the search user interface **210** includes a text entry box **208** that can receive alphanumeric input (e.g., from a keyboard) from the user. In some examples, the search user interface **210** can include additional interactive features, such as a preview window **215**. As described with some examples, the text entry box **208** can receive (i) search input to identify a plugin, and (ii) plugin input to enable the user to specify parametric values for running the identified plugin from the search user interface **210**. The preview window **215** can display entries for a search result (e.g., desired plugin of the user). As an addition or alternative, the preview window **215** can display prompts or predetermined plugin inputs which the user can select. Still further, in other variations, the search user interface **210** can include other types of interfaces and interactive elements.

(61) Plugin Library

(62) In examples, the plugin library **250** includes program files (e.g., executable files) which can execute at the selection of an end user in connection with the end user utilizing the interactive system **100** to create and/or update a design on a canvas. The plugins can be created by developers, including third-parties to a proprietor of the interactive system **100**. In examples, each plugin can be executable at the option of a user to implement a process separate from the functionality of the interactive system **100**. Accordingly, the plugins stored with the plugin library **250** can provide additional or enhanced functionality for use with interactive system **100**.

(63) In examples, a developer can interact with the plugin management system **200** to store a plugin file **255** (or set of files that are used at time of execution) with the plugin library **250**. The plugin files **255** can include one or more executable files for the plugin, as well as for plugin execution logic **245**. The plugin execution logic **245** can include code, program(s), programmatic processes and/or data which are accessed by the plugin management system **200** during a plugin search process, to prompt or facilitate the user in providing plugin input that is appropriately structured, formatted and/or sequenced.

(64) Use of Plugin Execution Logic

(65) In some implementations, the plugin execution logic **245** includes metadata specified by the developer, where the metadata includes parametric values that correlates to plugin inputs that a user can provide in connection with execution of the plugin. As an addition or alternative, the plugin execution logic **245** can be executed or otherwise utilized by the plugin interface component **230** of the system **200** to alter a configuration of the search user interface **210**. For example, as described in greater detail, the system **200** can utilize the plugin execution logic **245** to identify prompts **249** that appear in the text entry box **208** during a plugin search and execution process. In examples, a prompt **249** corresponds to a visual cue, such as a textual cue (e.g., a word) that indicates a type, format, value (or range of value) or other input characteristic which the user should enter as plugin input, in order for the plugin to successfully execute as intended by the user. As described below, in some examples, a prompt **249** can be a textual element within the text box entry **208**. In other examples, prompts **249** can appear as, for example, entries of the preview **215**, where each entry corresponds to an option that the user can make in order to specify a corresponding parametric value.

(66) The plugin execution logic **245** can be associated with prompts **249** and/or other interactive features that enable the user to provide specific parametric values to execute the selected plugin. In examples, the prompts **249** are formatted to appear as text-based suggestions with the search interface (e.g., within the text field of the search user interface **210**) to prompt the user for a specific type of value (e.g., a selection of one option from a closed set of options, a numeric value that indicates a range, a free form alphanumeric string, etc.). In some examples, the plugin execution logic **245** identifies prompts **249** that are displayed to the user upon the user performing a search that identifies the particular plugin. The user can then utilize the search user interface **210** to (i) perform a search to identify a plugin of interest, (ii) specify parametric values in accordance

with a sequence, format and/or structure to enable execution of the identified plugin, and (iii) trigger execution of the plugin using the specified parametric values. In some examples, the plugin execution logic **245** can also specify a set of default (or predetermined) parametric values that can be utilized to execute the plugin in absence of user input.

(67) In variations, the plugin execution logic **245** is distributed, so that the plugin execution logic **245** is stored in whole or in part at a developer-specified network location, such as with a developer or third-party service. For example, the plugin files **255** can include a pointer (e.g., URL or link) that includes (i) a pointer (e.g., URL, link) to a remote network resource or location, and/or (ii) other data for accessing developer specified files at the specified location. The portion of the plugin execution logic **245** that is stored with the developer-specified location (e.g., such as provided by a third-party service) can include, for example, instructions executable by the plugin interface component **230**, metadata that associates parametric values for the plugin with user input, prompts and/or other data for configuring the search user interface **210** and/or generating a separate input interface for the plugin.

(68) As an addition or variation, the plugin execution logic **245** can provide for the system **200** to utilize other types of visual elements and/or configurations that alter the appearance or functionality of the text entry box **208** during a plugin search and execution process. Still further, the plugin execution logic **245** can be executed by the system **200** to create a separate input interface for the user during a plugin search and execution process where the corresponding plugin is identified.

(69) In more detail, examples provide that the system **200** accesses and utilizes the plugin execution logic **245** to configure the search user interface **210** for a selected plugin, and to further enable execution of the plugin in a manner intended by the user. Accordingly, the plugin execution logic **245** is specific to the associated plugin. In some implementations, the plugin execution logic **245** can be provided as, for example, a data file that is uploaded by a developer through the developer interface **260**. In variations, the plugin execution logic **245** can be provided by a developer through an interactive aspect of the developer interface **260**. For example, the developer interface **260** can enable the developer to specify prompts **249** for the developer's plugin, where each prompt **249** identifies or indicates a possible parametric value for the user to enter or select. Additionally, the developer interface **260** enables the user to specify rules, conditions or other logic for when associated prompts **249** are to be rendered to the user (e.g., displayed within the text entry box **208**). By way of example, the conditions and rules specify, for example, the sequence of timing of when specific prompts **249** can appear (e.g., an initial prompt, a subsequent prompt), the logical relationship of one prompt to another prompt (e.g., a set of prompts are dependent on the input provided to another prompt, etc.), and/or other conditions for which a given prompt may appear in the text entry box **208**.

(70) As an addition or variation, for a given plugin that is selected during a plugin search and execution process, the system **200** can implement a decision flow by which a user enters a series of inputs based on a logical schema of the plugin execution logic **245**. As an addition or alternative, the system **200** can access the plugin execution logic **245** to provide prompt the user to make selection from multiple possible preselected parametric values. In such examples, the selections, sequence or timing and/or condition(s) under which the preselected parametric values are displayed to the user can be determined by the plugin execution logic **245**.

(71) Search Index

(72) The search index **240** can associate descriptive terms with individual plugins of the plugin library **250**. By way of example, the plugin files **255** can include search terms, such as descriptive terms that facilitate identification of the plugin through a search process. The descriptive terms can include terms that are characteristic to the plugin to facilitate identification of the plugin from other plugins of the plugin executable library **250**. The descriptive terms can also include parameters terms that are used by a respective plugin during the plugin's execution. and execution process.

(73) The search index **240** can be maintained separate or independent of the plugin library **250**. For example, the search index **240** can be downloaded and local on a user computing device, while the plugin library **250** is available to the user computing device over a network. The search user interface **210** can include text entry box **208** to receive alphanumeric input from the user, and the search component **220** implements a search process to identify matching entries using the search index **240**. The search component **220** can return a set of results (search results **211**) to the search user interface **210**. The search results **211** can include one or multiple entries which are displayed or presented to the user by the search user interface **210**, with each entry identifying a matching item (e.g., plugin) from the search index **240**.

(74) In some examples, the search index **240** can also include data that identifies commands, files, objects and/or data items that can be used by the interactive system **100**. In this way, the plugin management system **200** can enable the user to perform a universal search, where the search component **220** uses alphanumeric search input received via search user interface **210** to perform a search against search index **240**. The search component **220** can identify a search result **211** comprising entries, with each entry corresponding to a plugin, command, file and/or data object. With selection of a plugin from the search result (e.g., user selects TAB or ENTER), the system **200** receives subsequent alphanumeric entry as plugin input for use in execution of the identified plugin.

(75) Multimode Operation to Receive Search and Plugin Input

(76) According to examples, the user interacts with the search user interface **210** to provide (i) a first set of inputs (“search input **209**”) to select a plugin, and (ii) a second set of inputs (“plugin input **207**”) to specify parametric values for use during execution of the plugin. According to examples, the plugin management system **200** implements multiple modes in connection with receiving user inputs (e.g., search user input **209** and plugin user input **207**) via the search user interface **210**. In particular, the plugin management system **200** implements a search mode in which a first set of user inputs (i.e., search input **209**) is used to identify a particular plugin. Once the user selects a desired plugin, the system **200** switches to a plugin input mode, where a second set of inputs (“plugin input **207**”) are used as input for execution of the desired plugin. Each of the modes are described in greater detail below.

(77) Search Mode

(78) In the search mode, the search component **220** receives the search input **209** and performs a search using search index **240** to identify the search results **211**, where the search result **211** includes one or more plugins from the plugin library **250**. The user can select one of the plugins from the search result **211** using, for example, a keyboard. The search user interface **210** can generate preview window **215** to display entries of the search result **211**. In examples, the user can interact with the preview window **215** to select one of multiple entries of the search result **211**. In some examples, the user input can be provided through keyboard entry. For example, the user can use a keyboard to enter one or more character strings in the text entry box **208** to generate the search result **211**, which is displayed to the user via the preview **215**. The user can use arrow keys or other directional input (including pointer) to select a desired plugin from a portion of the search result **211** displayed in the preview **215**. The user can then press TAB on the user keyboard to select the desired plugin (or other entry displayed through the preview **215**).

(79) In variations, the search component **220** implements alternative search processes to identify matching entries from the search index **240**. For example, the search component **220** can (i) match search input **209** with any part of a descriptive term of a plugin stored with the search index **240**, (ii) match search input **209** exactly to a descriptive term, (iii) match search input **209** with the beginning portion of a term, and/or (iv) utilize substitutions or “fuzzy” logic for a search term or portion thereof.

(80) Still further, the search user interface **210** and search component **220** can combine to implement an incremental search. The search component **220** can perform a search of the search

index **240** starting with a first character that the user enters with the search user interface **210**, and subsequent character input can cause the search results in narrow. The search user interface **210** can provide preview **215** with each character input that the user enters, with the matching entries displayed in the preview **215** being reduced with each additional alphanumeric entry.

(81) In other examples, the search component **220** can use the detected alphanumeric entry to search the search index **240** for matching items that include, for example, plugins, commands, files, data objects and other data items. The search component **220** can return a search result **211** that identifies one or more entries, where each entry corresponds to a matching item (e.g., plugin, command, file, data object or other data item).

(82) In some examples, once the user selects a desired plugin, the search component **220** causes an identifier of the selected plugin to render as an object in the text entry box **208**. The completion of the search process can transition the system **200** from the search mode to the plugin input mode.

(83) Plugin Input Mode

(84) In response to user selection of a plugin, the plugin management system **200** can switch to a plugin input mode. In the plugin input mode, plugin interface component **230** accesses the plugin execution logic **245** of the selected plugin from the plugin library **250**, and configures the search user interface **210** based on the plugin execution logic **245** of the selected plugin. In an aspect, the search user interface **210** is configured to provide prompts **249** that enable the user to specify parametric values for executing the selected plugin through the search user interface **210**. In examples, the plugin interface component **230** displays prompts **249** within the text entry box **208** of the search user interface **210**, where the prompts **249** are specified by the plugin execution logic **245** of the selected plugin. The prompts **249** can specify or indicate parametric values that the user can provide as plugin input **207** for use in execution of the selected plugin. Still further, as described with some examples, the plugin interface component **230** can implement plugin execution logic **245** to generate other interactive features to facilitate the user in providing plugin input that identifies parametric values for execution of the selected plugin.

(85) From the user's perspective, selection of a desired plugin results in the search interface being reconfigured to indicate prompts or other inputs which the user can make to specify parametric values for the desired plugin. Subsequent input entered by the user is received as plugin input **207** by the plugin interface component **230**. In some implementations, the plugin input **207** can also be in the form of alphanumeric input that correspond to parametric values for running the selected plugin. In this way, the search input **209** and plugin input **207** can be entered by the user as a successive or continuous set of inputs which the user provides to search and execute the desired plugin. Further, the plugin input **207** may be provided by the user in connection with an identified plugin, and responsive to a set of prompts **249** (or other interactive elements specified through the plugin execution logic **245**). In examples, the set of prompts **249** that are displayed to the user are determined from the plugin execution logic **245** of the matching plugin.

(86) Accordingly, in the plugin input mode, the plugin interface component **230** can access and implement the plugin execution logic **245** from the plugin library **250** to (i) identify what prompts to display initially using the search user interface **210**, (ii) identify input provided by the user through the search user interface **210** that specifies one or more parametric values for a corresponding set of parameter terms identified by a displayed prompt, and (iii) implement decision logic to determine which prompt to display (if any) after parametric values for a given set of displayed parameter terms are received. In some examples, the prompts **249** displayed to the user visually indicate or guide the user to enter one or more parametric values for the plugin in a particular format (e.g., numeric value, numeric value in a range, selection of one term from finite set of possible terms, etc.), structure (e.g., sequence of two parametric values), and/or type (e.g., selection input versus alphanumeric input). Once the plugin input is complete, the user can enter input (e.g., the user presses ENTER in keyboard) to trigger execution of the selected plugin, using parametric values specified by the plugin input. In this way, the search user interface **210**, search

component **220** and plugin interface component **230** combine to enable the user to enter successive inputs that are used to search and execute a desired plugin in an intended manner of the user.

(87) In some variations, the plugin execution logic **245** is stored in whole or in part at a developer-specified network location, such as with a developer or third-party service. The plugin interface component **230** can be configured to access the developer-specified network location to retrieve and implement the plugin execution logic **245**. In some implementations, the plugin files **255** of the developer include a link, URL or other identifier to the network location, and/or as well as other data (e.g., encryption key, credential data, etc.) to enable the plugin interface component **230** to utilize the plugin execution logic **245** stored at the network location.

(88) In some examples, the plugin interface component **230** can implement the plugin input mode by displaying multiple entries for a selected plugin, where each entry identifies one or more parametric values for use with the selected plugin. In some variations, the user can provide directional or scrolling input to change entries. Additionally, the user may interact with the search user interface **210** to change or modify the parametric value associated with a particular entry displayed in the preview **215**.

(89) As further described in examples, when trigger input **219** is provided by the user via the search user interface **210**, the plugin interface component **230** can trigger execution of the corresponding plugin from the plugin library **250**. The selected plugin is then deployed while the user is using the interactive system **100**, in connection with content rendered on a canvas.

(90) In some examples, the plugin interface component **230** can respond to selection of an entry that specifies a parameter term and parametric value by triggering the corresponding plugin to execute using the parameter and specified parametric value of the selected entry. Still further, in other variations, the plugin interface component **230** can respond to selection of an entry that specifies a parameter term by triggering the corresponding plugin to execute using the parameter specified in the selected entry and a default (or predetermined) parametric value. The plugin can then execute to, for example, provide additional or enhanced functionality for the user in connection with the user utilizing the interactive system **100**. For example, a triggered plugin can modify the content **125** that is rendered on the canvas **122** using functionality provided by the triggered plugin. In some examples, the plugin interface component **230** stops running a selected plugin once the plugin performs desired operations using the specified or default parametric values.

(91) Input Validation Logic

(92) In some examples, the plugin interface component **230** implements validation logic to ensure user input is valid for the selected plugin. In some examples, the plugin interface component **230** can implement the validation logic to check whether the input provided by the user matches a format or value specified by a corresponding prompt **249**, or by metadata that is associated with the prompt **249**. The plugin interface component **230** can for example, match alphanumeric entry provided by the user to one of the prompts **249**, or to metadata associated with the current prompt **249** that the user is providing input for.

(93) In examples, the plugin interface component **230** implements the validation check in response to each alphanumeric entry of the user. If the validation check determines that one entry in the string of entries of a parametric term is invalid, the plugin interface component **230** can implement any one of a variety of corrective actions. In one implementation, the plugin interface component **230** can ignore the input that failed the validation check. In a variation, the input can be displayed, but a respective string that contains the invalid character is visually indicated as being invalid. For example, when a string for a parametric term is completed and matches a valid input, the plugin interface component **230** can display the string as an object (e.g., object representing the parametric term which the user entered). But if the user completes the string and one of the characters is deemed invalid, then the string is displayed in non-object form (e.g., as a string of characters). Still further, an invalid character or string can be highlighted, colored or otherwise visually marked to be invalid. Still further, in other variations, the plugin interface component **230** can autocorrect an

invalid entry for the user, using auto-correct logic. Still further, the plugin interface component **230** can access the plugin execution logic **245** to determine the manner in which an invalid string or input is to be handled.

(94) As an illustrative example, a desired plugin may enable the user to identify an image of a fish, and the prompt for the plugin may specify “color”. The plugin interface component **230** can perform the validation check after each alphanumeric entry of the user. In a scenario where the user enters “b”, “l” and “o”, the plugin interface component **230** may determine the string as being invalid when the “o” is received. The user may go back and correct the string (e.g., enter “u” and “e”) or the user may determine that he or she accidentally selected the wrong plugin (e.g., the user was entering “blonde” for hair. In such an example, once the plugin interface component **230** determines that the string the user is entering for the plugin input **207** does not match a valid response, the plugin interface component **230** can provide a visual indication that the entered string is not valid for the particular plugin.

(95) In some examples, the plugin interface component **230** implements validation logic that is specified by the plugin execution logic **245** of the selected plugin. In variations, the plugin interface component **230** includes validation logic that is implemented independent of the plugin execution logic **245**.

(96) Rich Media and Functional Plugin Interfaces

(97) According to some examples, the plugin interface component **230** accesses and executes plugin execution logic **245** of a selected plugin from the plugin library **250** to render input interfaces that are separate from the search user interface **210**. A developer can structure the plugin execution logic **245** to enable the plugin interface component **230** to execute code that is provided as part of the plugin execution logic **245**, from which a separate plugin input interface can be rendered to the user. In some implementations, the plugin interface component **230** accesses the plugin execution logic **245** to generate a separate plugin input interface once the corresponding plugin is selected by the user. The plugin input interface can be rendered concurrently with, for example, the text entry box **208** of the search user interface **210**. Through the plugin input interface, the user can be prompted or otherwise guided to provide plugin inputs that are used by the selected plugin during its execution.

(98) In examples, the separate interface can include or provide rich media. In some implementations, the plugin interface component **230** utilizes an iframe when executing plugin execution logic **245** to render rich and/or dynamic input interfaces for the selected plugin. For example, the plugin interface component **230** can execute the plugin execution logic **245** to generate an interface that enables the user to specify an object (e.g., picture), and to dynamically provide input based on the user's prior input. As an addition or variation, the plugin interface component **230** can access the plugin execution logic **245** to select images or other media to render for prompting user input (e.g., display matching image results to a selected image and prompt the user for selection).

(99) Furthermore, the plugin interface component **230** can render the plugin input interface to be dynamic. For example, the plugin execution logic **245** can specify input features that are conditional or sequenced, so that a prior user plugin input determines the input feature that is rendered to the user. For example, the plugin interface component **230** can implement the plugin execution logic **245** to generate input interface elements that are dependent on the user's selection or prior input. As an illustrative example, the plugin interface component **230** can generate a plugin input interface that displays a shape having characteristics that are determined by a user's selection (e.g., a user may select a shape from multiple possible shapes), or alternatively by a user's prior input (e.g., the user previously selected a rectangular shape, and the user is prompted to select from a rectangular shape). Through the dynamic interface provided by the plugin interface component **230**, the user can enter a series of inputs that can in turn, be used in the execution of the selected plugin to create a flow chart of linked shapes on the canvas.

(100) Still further, the plugin input interface generated by the plugin interface component **230** can enable the user to provide inputs of various types, and the features provided with the plugin interface can be based in part on the plugin input the user previously entered. Additionally, the plugin interface component **230** can, based on the plugin execution logic **245**, generate the separate plugin interface based on plugin input the user initially enters via the text entry box **208**. Thus, for example, the plugin interface component **230** can implement the plugin execution logic **245** to selectively generate the plugin interface separate from the text entry box **208**. In this way, the type, format and logic used to prompt the user for input and/or receive input from the user can be determined by the plugin execution logic **245** of the selected plugin, as implemented by the plugin interface component **230**.

(101) Hidden Metadata

(102) Still further, in some examples, the system **200** enables a developer to structure the plugin execution logic **245** of a plugin to include metadata that specify parametric values that coincide with plugin input provided by a user via the search user interface **210**. The metadata may be hidden or transparent to the user. A developer can structure the plugin execution logic **245** to include multiple sets of metadata, where each metadata set associates a given set of parametric values with a corresponding plug-in input of the user. In response to receiving plugin input via the search user interface **210**, the plugin interface component **230** identifies the metadata set that is associated with the received plugin input. In this way, the plugin interface component **230** can trigger the selected plugin to execute using parametric values specified by plugin inputs received via the search user interface **210** and/or parametric values specified with metadata set(s) that are associated with the plugin input(s).

(103) By way of example, the plugin interface component **230** can implement the plugin execution logic **245** of a given plugin to display multiple options for the user, where each option corresponds to a possible plugin input that the user can provide via the search user interface **210**. For example, based on the plugin execution logic **245**, the plugin interface component **230** can provide multiple entries in the preview window **215**, where each entry corresponds to a possible plugin input. The user can scroll or otherwise provide directional input in the preview window **215** to select one of the multiple possible plugin inputs. In examples, the plugin interface component **230** accesses plugin execution logic **245** of the selected plugin to identify a metadata set that is associated with the selected plugin input. The plugin interface component **230** can subsequently trigger the selected plugin to execute using parametric values specified in the identified metadata set.

(104) As an illustrative example, the user plugin input can include a selection of an iconic representation of an object, and the plugin interface component **230** can access the plugin execution logic **245** to identify a metadata set that is associated with a particular icon. The plugin interface component **230** can trigger the identified plugin to execute using, for example, a unique identifier that is specified by the associated metadata and which distinguishes the icon from other icons of an icon library used by the plugin.

(105) As another illustrative example, the plugin interface component **230** can access the plugin execution logic **245** to identify a metadata set associated with a particular user selection of an icon, where the parametric values identified by the associated metadata set correspond to one or more attributes reflected in the representation of the icon. For example, a user selection of an icon representing an adult male can be associated with metadata that correspond to attributes of “male” and “age 30-45 years old”.

(106) As another illustrative example, the plugin interface component **230** can execute plugin execution logic **245** of the selected plugin to render a map interface. For example, the plugin interface component **230** can initially provide prompts **249** for the user to enter a geographic region in the text entry box **208**, and the plugin interface component **230** can further execute the plugin execution logic **245** to generate a map interface for the geographic region specified by the user's input. The plugin interface component **230** can render the map interface to enable the user to

interact by providing an input. For example, the user can enter (e.g., using a keyboard) an address or location identifier (e.g., “Tenderloin” or city neighborhood), and the map interface generated by the plugin interface component **230** can be dynamically responsive to enlarge and center about the address or identifier the user provided. As an addition or variation, the user can provide a directional or pointer input (e.g., via a mouse) on a selected location that represents a corresponding geographic location within the represented geographic region. When the user provides plugin input to select the location within the represented geographic region, the plugin interface component **230** can identify a metadata set for the selected location of the user, where the identified metadata set identifies, for example, the geographical coordinates (longitude and latitude) of the selected location.

(107) Historical Information

(108) In examples, the plugin management system **200** can record the user's activity through the search user interface **210**. The plugin management system **200** can utilize the historical information to (i) identify plugins which the user is likely interested in, and (ii) values that the user previously specified in connection with each of the identified plugins. The historical information can, for example, be used to identify plugins and/or values for the plugins, based on the user's most recent or most frequent selection. For example, the historical information can be used to indicate the plugin identifier that the user most recently or frequently selected. Additionally, the historical information can be used to indicate the values which the user most recently or frequently specified with the respective plugin identifier.

(109) Methodology

(110) FIG. **3A** describes an example method for providing a plugin library, according to one or more embodiments. FIG. **3B** illustrates another example method for enabling plugin execution, according to one or more embodiments. Examples such as described with FIG. **3A** and FIG. **3B** can be implemented by a plugin management system such as described with an example of FIG. **2**. Further, examples such as described with FIG. **3A** and FIG. **3B** can be implemented in connection with an integrated system or platform that utilized by a user, such as described with FIG. **1A** through FIG. **1C**. Accordingly, reference is made to elements of FIG. **1A** to FIG. **1C** and FIG. **2** for purpose of illustrating a suitable component for performing a step or sub-step being described.

(111) With reference to FIG. **3A**, the plugin management system **200** includes plugin library **250** which maintains a plurality of plugins that are executable in connection with individual users operating the interactive system **100** (**310**). The plugin library **250** associates each plugin with one or more descriptive terms. In examples, the descriptive term(s) can be identified by the developer, and can include identifiers and/or descriptors about the functionality provided by the plugin. The plugin management system **200** can include developer interface **260** to enable a developer to upload a plugin, as well as descriptive terms and descriptors of the plugin.

(112) The plugin management system **200** can provide a search user interface to enable the user to search for a plugin. For example, the search user interface **210** can be provided in connection with the interactive system **100** being implemented on a user computing device (**320**). The search user interface **210** can be in the form of, for example, a text entry box that receives alphanumeric entries from a user operating a keyboard.

(113) In response to receiving the search term, the plugin management system **200** performs a search operation to match the search term with one or more plugins, using, for example, descriptive terms specified by the developer (**330**). In receiving the search term, the plugin management system **200** can detect a first set of input for the user, where the first set of input corresponds to the search term. In some implementations, the plugin management system **200** can perform a search based on detecting inputs of the user that correspond to individual characters of the search term. Thus, the detected inputs of the user can narrow a resulting search result.

(114) The plugin management system **200** can return a search result that includes one or more entries which identify one or more plugins stored with the plugin library **250** (**340**). The user can

provide a trigger input to cause execution of the plugin represented by a selected entry of the search result (350). In examples, the user can execute the plugin via alphanumeric input provided through the search user interface 210. For example, the user can utilize, for example, a keyboard to provide (i) alphanumeric input that specifies parametric values for execution of the selected plugin, and (ii) trigger input (e.g., via ENTER key) to trigger execution of the selected plugin.

(115) With reference to an example of FIG. 3B, plugin management system 200 maintains a data store that identifies a plurality of plugins (360). As described with other examples, the plugin management system 200 can be implemented as part of an integrated system or platform to provide a user with a canvas on which the user can generate and update content, such as designs, whiteboards, presentations, web pages and other types of content. The library of plugins can be maintained by the plugin management system 200 as part of a network service provided by the network computing system 150. Alternatively, the library of plugins can be stored locally on the user computing device, or distributed between the user computing device and the network computing system 150.

(116) In examples, the plugin management system 200 determines a plugin based on a first set of input provided by a user of a computing device (370). In examples, the plugin management system provides a search interface to receive the inputs of the user. The inputs of the first set can include, for example, the alphanumeric entries that are entered through, for example, a text entry box of the search interface. The plugin management system 200 identifies a plugin from the library of plugins, based on the first set of inputs. The plugin management system 200 implements one or more search operations (responsive to inputs received by the user) to match the user input(s) with a matching plugin from the library of plugins.

(117) Further, the plugin management system 200 determines a set of parametric values for the identified plugin based on a second set of input provided by the user (380). In some examples, the search user interface 210 can be used to display prompts that indicate a type or range of parametric values for the determined plugin. For example, the search user interface 210 can include a text entry box 208 or a preview panel 215 where the prompts can be provided. The user can provide input to select or otherwise specify values for the prompts.

(118) In examples, the plugin management system 200 triggers execution of the identified plugin using the set of parametric values in response to detecting a first trigger input provided by the user (390). Among other advantages, a method such as described by examples simplify the interactions required from the user to identify and execute a desired plugin. For example, the user can use a common input device (e.g., keyboard) to search and execute a desired plugin, and further, without the user having a priori knowledge or familiarity of how the plugin works. Rather, the search user interface 210 can provide a guide as to how the plugin is to execute.

EXAMPLES

(119) FIG. 4A through FIG. 4D illustrate an example of a search user interface for selecting and triggering execution of a desired plugin, according to one or more embodiments. FIG. 4E and FIG. 4F illustrate another example of a search user interface for selecting and triggering execution of plugins, according to one or more embodiments. FIG. 4G illustrates another example of a search interface that enables universal search, including plugin selection and execution, according to one or more embodiments. FIG. 4H through FIG. 4J illustrate another example of a search interface to invoke additional functional interfaces for interacting with plugins, according to one or more embodiments. FIG. 4K through FIG. 4O illustrate another example of a search interface that utilizes historical information, according to one or more embodiments. In describing examples of FIG. 4A to FIG. 4D, FIG. 4E to FIG. 4F, FIG. 4G, FIG. 4H through FIG. 4J, and FIG. 4K through FIG. 4O, reference is made to elements of FIG. 1 and FIG. 2 for purpose of illustrating components for implementing functionality as described. Accordingly, examples a search interface 400 as shown and described by FIG. 4A through FIG. 4O can be provided on a display 402 of a user computing device by interactive system 100. Further, the user can interact with search user

interface **400** in a variety of ways, such as through keyboard, pointer (e.g., mouse) and/or touch input. The search user interface **400** is provided on a display **402** of a user computing device, in connection with a user utilizing the interactive system **100** to create, edit or collaborate on content rendered on a canvas. In examples, the interactive system **100** generate the **400** in response to a designated user input, such as a short-cut operation (e.g., provided through keyboard entry), soft-feature selection (e.g., iconic representation of feature) or pull down menu. Further, when the search interface **400** is provided, subsequent input received through the search interface **400** can invoke components of the plugin management system **200**, as described with examples of FIG. 2. (120) With reference to an example of FIG. 4A, the search interface **400** is provided on the display **402** to receive a first set of inputs **412** in a text entry box **410**. The first set of inputs **412** can include alphanumeric characters (e.g., “ui”) provided by, for example, the user through use of a keyboard or other alphanumeric entry mechanism. The search component **220** can match the first set of inputs **412** with descriptive terms that are maintained with the search index **240**. The search component **220** can use the first set of inputs **412** to display a preview **414** that includes an identifier **418** (e.g., “UI Faces”) to a matching plugin. The preview **414** can further include a description **419** of the identified plugin, which may be provided by the developer via, for example, the developer interface **260**. The user can select the plugin from the preview **414** and/or through additional input. For example, the user may provide additional input through the keyboard (e.g., by pressing TAB) to signify selection of a particular entry that is displayed in preview **414** as part of the search result.

(121) In some examples, one or more prompts **422** are displayed to the user in connection with a plugin that appears in the preview **414** in a pre-selected state (e.g., plugin identifier is the only entry identified in the preview as part of the search result, or the plugin identifier is highlighted in the preview). The search user interface **210** and/or search component **220** can retrieve one or more prompts **422** specified as part of the plugin execution logic **245** from the plugin data library **250**. The prompts **422** can be displayed within search entry box **410**. For example, one or more prompts **422** can be displayed as lightened text within the search entry box **410**. The prompts **422** can correspond to parameter terms of the pre-selected plugin. The initial prompts **422**, the sequence in which multiple prompts **422** are displayed, the type of input required for the input to be accepted and run as part of the plugin, and whether or not default values can be predetermined and associated with the particular parameter term are examples of variations that can be specified by the developer with the plugin execution logic **245**. Thus, the plugin execution logic **245** can specify parametric values by parameter, type and configuration, to facilitate the user to trigger execution of the desired plugin through use of the keyboard.

(122) With reference to an example of FIG. 4B, the selected plugin identifier **418** is displayed as an object in the search box interface **410**. For example, the user can select the plugin identifier **418** when it is shown in the preview **414** by pressing TAB, causing selection of the first (or only) entry of the preview **414**, or by clicking the plugin identifier **418** when it appears in the preview **414**. When selected, the plugin identifier **418** appears as an object, with pre-determined prompts **422** appearing adjacent to the object in the text entry box. As described in FIG. 4A, the pre-determined prompts **422** can be specified with the plugin execution logic **245** associated with the particular plugin. Thus, the appearance of the plugin identifier **418**, as well as the prompts **422** and other behavior or characteristics of the plugin can be made plugin specific.

(123) With further reference to an example of FIG. 4B, the preview **414** provided through interaction with the search interface **400** can be changed to display possible values for the displayed prompts **422**. In the illustrative example shown, prompts **422** are sequenced so that the first parametric value the user provides is for gender, and the second parametric value is for age. In connection with the first parametric value, preview **414** displays a set of possible values **424** from which the user can make a selection. For example, the user can provide keyboard input (e.g., directional input, TAB, or other combination) or other types of input to select one of the possible

values **424** as a selection of a parametric value for the given parameter term represented by the first prompt **422**.

(124) As illustrated by an example of FIG. **4C**, when selection of the possible values is made for the first prompt **422**, the text box interface **410** of search interface **400** displays possible parametric values **428** specified by the plugin execution logic **245** for the parameter term represented by the second prompt **422**. The preview **414** is updated to display indicators of possible parametric values **428** for the second prompt **422** (e.g., “age”). The search box interface **410** can display the selected parametric value **424** for the first prompt **422**, adjacent to the selected plugin identifier **418**. Each of the possible parametric values **428** for the second prompt **422** can correspond to a range of age values (40-60), or alternatively, to a particular age value (e.g., “45”, “any”), depending on the input the user specifies. The user can specify the particular value for the second prompt **422** using a keyboard or other input device. In this way, the possible input values, value types and type of input which may be accepted as used input can be specified by the developer with the plugin execution logic **245**.

(125) FIG. **4D** furthers the example illustrated by FIG. **4A** through FIG. **4C**, to illustrate the case where the user has selected a plugin, and also specified one or more parametric values indicated by prompts **422** (FIG. **4C**). In such case, the search box interface **410** displays the parametric values **424** for the parameter terms represented by the respective prompts. The user can provide trigger input (e.g., ENTER on the user keyboard) to trigger execution of the selected plugin, using the specified parametric values during the execution of the plugin. The plugin executes to perform an operation that changes, for example, the design under edit, the canvas or the design panel.

(126) With reference to an example of FIG. **4E** and FIG. **4F**, the search interface **400** can be provided on the display **402** to implement a preview **414** that include iconic or graphical representations of an output of the plugin. As described, the appearance and behavior of the preview **414** in connection with the rendering of plugin parameters and values can be specific to the plugin. Upon the user providing a first set of inputs to select a plugin (e.g., “Material Design Icons”), the preview **414** displays possible parametric values **424** for the parameter term indicated by a first prompt **422** (e.g., “Icon Name”). The possible values **428** which the user can select can be displayed in scrollable form with the preview **414**. Thus, for example, the developer can configure the preview **414** to enable scrolling or directional input (e.g., using directional arrows of the keyboard) to accommodate the number of possible choices the user may have for the parametric value of the parameter term represented by any of the prompts **422**. Additionally, in some examples, each possible value **428** is associated with hidden metadata that can identify specific parametric values for use at the time the plugin executes.

(127) In an example shown, the parametric values are displayed with the preview **414** in alphabetical order. However, the plugin execution logic **245** can specify alternative orders, including, for example, specifying the most recent or most commonly used parametric value to appear first in the list provided with preview **414**. Once the parametric value for the first prompt **422** is selected, possible parametric values **428** for the parameter term represented by the second prompt **422** can be displayed as part of the preview **414**. Likewise, the selected parametric value **424** for the first prompt can be displayed in the search box interface **410**.

(128) With reference to FIG. **4G**, examples provide that the plugin management system can implement the search interface **400** to include functionality that extends beyond plugin search and execution. As illustrated by FIG. **4G**, the search interface **400** can be implemented in, for example, alternative modes, including a universal search mode in addition to a plugin or plugin input mode. In particular, the search interface **400** can be implemented as part of the universal search feature of the interactive system **100**. For example, search index **240** can maintain descriptive terms and identifies for plugins, as well as commands, data objects, files, tools and other types of items which can be used with the interactive system **100**. Further, the search index **240** can include local or account specific index or search resources, such as file names, templates, objects, etc. In such

examples, a first set of alphanumeric input **442** can generate a search result **445** that is displayed as part of preview **414**, with individual entries **441** identifying plugins, as well as commands, data objects, files, tools, etc. The preview **414** can be scrollable using, for example, directional input provided through keyboard. Alternatively, entries **448** which represent plugins can be selected using a shortcut key combination, such as shown by indicators **449**.

(129) With examples such as illustrated by FIG. 4A to FIG. 4D, FIG. 4E to FIG. 4F, and FIG. 4G, the type of input which a user can provide can be specified as part of the plugin execution logic **245**. For example, to specify a parametric value for a selected plugin, the plugin execution logic **245** can enable the user to provide input that corresponds to directional input (e.g., through directional keys) and selection input (e.g., pressing ENTER or TAB). Alternatively, the plugin execution logic **245** can specify the type of input that can be received as being numeric entry, but not text entry. As another example, the plugin execution logic **245** can enable the user to specify alphanumeric input so long as the alphanumeric input as entered in succession matches with one of the options displayed in the preview. As still another example, the plugin execution logic **245** can enable the user to specify free-form alphanumeric input which can be received as a parametric value of a corresponding parameter term, free of predetermined format or structure. Depending on implementation, the plugin execution logic **245** can thus enable a developer to restrict (or not) the input the user can provide by data type, format, structure or other consideration. Furthermore, the plugin execution logic **245** can specify functional behavior or other configurations for how the preview **414** is presented to enable the user to select parametric values for their plugin.

(130) With reference to examples illustrated by FIG. 4H through FIG. 4J, the search interface **400** can be used to invoke additional functional interfaces for interacting with plugins. In examples, the plugin management system **200** can implement plugin execution logic **245** to generate separate functional interfaces for enabling the user to enter parametric values for execution of the plugin. In an example illustrated by FIG. 4H, the user interacts with the text entry box **410** of search interface **400** to specify a search entry (e.g., “Colo”). The search component **220** implements a search operation to identify a matching plugin (e.g., “Color Selector”).

(131) In examples illustrated by FIG. 4I and FIG. 4J, once the plugin is identified, the text entry of the user is displayed as a plugin identifier **418**. For example, the plugin identifier **418** can be displayed within the text entry box **208** as an object. As described with other examples, the selection can coincide with the search interface **400** switching to a plugin input mode, to enable the user to specify parametric input for the identified plugin.

(132) In an example illustrated by FIG. 4I, the identification of the particular plugin can also coincide with a plugin input interface **460** being provided, separate from the search interface **400**. In example, the plugin input interface **460** can be implemented using plugin execution logic **245** associated with the identified plugin. For example, the plugin interface component **230** executes the plugin execution logic **245** of the identified plugin to generate the plugin interface **460** (e.g., color picker) for enabling the user to enter parametric input. The plugin interface **460** can be generated separate from the search interface **210**, such as in a region adjacent to the search interface **400**. Thus, for example, the plugin input interface **460 460** can be generated concurrently and/or in place of the search interface **400**. In examples, the plugin interface **460** can include rich media and graphics, such as implemented with an iframe, so as to enable a dynamic interface from which the user can indicate parametric values for use in execution of the plugin. Further, the plugin interface **460** can be dynamic. For example, the user can slide or otherwise manipulate an input feature **462, 464** (e.g., sliders) to select a desired value. Based on the input, the position of the input features **462, 464** can be changed. Metadata can be associated with the selected position of the input feature **462, 464**, where the metadata identifies one or more parametric values (e.g., selected hue for execution of the selected plugin). The selected plugin can then be executed with the parametric values specified by the user's manipulation of the input features.

(133) In a variation illustrated by FIG. 4J, the functional interface that can be provided with the

search interface **400** can be configured based on logic associated with the plugin. For example, the plugin interface component **230** can execute the plugin execution logic **245** to generate a second plugin interface **470** (e.g., color wheel) that enables dynamic user interactions. In example shown, the user can move a pointer **472** or other input mechanism (e.g., keyboard) over a color wheel or palette in order to make a selection of a desired hue. The plugin interface component **230** can associate the user's position selection with metadata that identifies the parametric values for the selected plugin. Once the user makes the selection, the user can trigger the plugin to execute with the identified parametric values.

(134) FIG. **4K** and FIG. **4L** illustrates another example in which the search interface **400** provides the user with plugin search and input history. As illustrated in FIG. **4K**, the plugin identifier **418** is displayed in the text box **410**, along with a prompt **422** that is pre-associated with the identified plugin. Further, preview panel **414** indicates possible or alternative values for the prompt **422**, from which the user can make selection (e.g., select pink).

(135) As illustrated by FIG. **4L**, when the user makes a selection of a possible value for the identified plugin, the user's selection is recorded and used again in a subsequent instance when the plugin is identified through the search interface **400**. In this way, the user's input for the value(s) of the prompt(s) **422** is recorded as historical information, and the historical information is used by the interactive system **100** to populate (or pre-populate) the value used for the prompt(s) **422** of the identified plugin. The value selection for the prompt **422** can be displayed in the text box **410** and confirmed by the user or replaced by another user selection.

(136) FIG. **4M** and FIG. **4N** illustrates another example where the search interface **400** utilizes historical information that coincides with calculated or inferred input, determined by the user's prior interaction with the plugin through the search interface **400**. In FIG. **4M**, the user interacts with the search interface **400** to enter characters for identifying a plugin (e.g., "calc") from a plugin library.

(137) In an example illustrated by FIG. **4N**, the user can enter (e.g., within the search interface **400**) parametric input, such as input for calculating a result. The PEL **245** associated with the identified plugin calculates or otherwise infers a determined value **444** from the input entered subsequent to the identification of the plugin. In an example shown, the value **444** is displayed in the preview panel **414**. In other examples, the value **444** is displayed adjacent to the search interface **400**. Still further, in other variations, the value **444** is a parametric input for use with execution of the identified plugin, and selection of the value **444** is used by the plugin to determine or generate another output from the plugin.

(138) In an example illustrated by FIG. **4O**, the user's prior interaction with the search interface **400** results in a history of the user's prior selections being displayed with the search interface **400**. In examples, the history **415** of the user's selection are displayed in the preview panel **414**. For example, the preview panel **414** can display the most recent plugin identifiers **418A**, **418B** in the preview panel **414**, along with the most recent parametric values **444A**, **444B** used or determined with the identified plugins. For example, the recent parametric values **444A**, **444B** of the identified plugins can include prior selections (e.g., "pink"), or determined values from prior parametric input (e.g., **4.4**). Selection of the plugin identifiers **418A**, **418B** may result in i) automatic population of the text entry box **410**, so that the selected plugin identifier **418A**, **418B** is displayed, along with the corresponding parametric value, or ii) automatic execution of the selected plugin, using the parametric values **444A**, **444B**.

(139) While examples illustrated by FIG. **4K** through FIG. **4O** illustrate use of historical information to determine recent selections, in variations, the historical information can determine most popular or most likely user selections.

(140) Network Computer System

(141) FIG. **5** illustrates a computer system on which one or more embodiments can be implemented. A computer system **500** can be implemented on, for example, a server or

combination of servers. For example, the computer system **500** may be implemented as the network computing system **150** of FIG. 1A through FIG. 1C, and further utilized by a plugin management system **200** of FIG. 2.

(142) In one implementation, the computer system **500** includes processing resources **510**, memory resources **520** (e.g., read-only memory (ROM) or random-access memory (RAM)), one or more instruction memory resources **540**, and a communication interface **550**. The computer system **500** includes at least one processor **510** for processing information stored with the memory resources **520**, such as provided by a random-access memory (RAM) or other dynamic storage device, for storing information and instructions which are executable by the processor **510**. The memory resources **520** may also be used to store temporary variables or other intermediate information during execution of instructions to be executed by the processor **510**.

(143) The communication interface **550** enables the computer system **500** to communicate with one or more user computing devices, over one or more networks (e.g., cellular network) through use of the network link **580** (wireless or a wire). Using the network link **580**, the computer system **500** can communicate with one or more computing devices, specialized devices and modules, and/or one or more servers.

(144) In examples, the processor **510** may execute service instructions **522**, stored with the memory resources **520**, in order to enable the network computing system to implement the network service **152** and operate as the network computing system **150** in examples such as described with FIG. 1A through FIG. 1C.

(145) The computer system **500** may also include additional memory resources (“instruction memory **540**”) for storing executable instruction sets (“interactive system instructions **545**”) which are embedded with web-pages and other web resources, to enable user computing devices to implement functionality such as described with the interactive system **100**.

(146) As such, examples described herein are related to the use of the computer system **500** for implementing the techniques described herein. According to an aspect, techniques are performed by the computer system **500** in response to the processor **510** executing one or more sequences of one or more instructions contained in the memory **520**. Such instructions may be read into the memory **520** from another machine-readable medium. Execution of the sequences of instructions contained in the memory **520** causes the processor **510** to perform the process steps described herein. In alternative implementations, hard-wired circuitry may be used in place of or in combination with software instructions to implement examples described herein. Thus, the examples described are not limited to any specific combination of hardware circuitry and software.

(147) User Computing Device

(148) FIG. 6 illustrates a user computing device for use with one or more examples, as described. In examples, a user computing device **600** can correspond to, for example, a workstation, a desktop computer, a laptop or other computer system having graphics processing capabilities that are suitable for enabling renderings of design interfaces and graphic design work. In variations, the user computing device **600** can correspond to a mobile computing device, such as a smartphone, tablet computer, laptop computer, VR or AR headset device, and the like.

(149) In examples, the computing device **600** includes a central or main processor **610**, a graphics processing unit **612**, memory resources **620**, and one or more communication ports **630**. The computing device **600** can use the main processor **610** and the memory resources **620** to store and launch a browser **625** or other web-based application. A user can operate the browser **625** to access a network site of the network service **152**, using the communication port **630**, where one or more web pages or other resources **605** for the network service **152** (see FIG. 1A through FIG. 1C and FIG. 2) can be downloaded. The web resources **605** can be stored in the active memory **624** (cache).

(150) As described by various examples, the processor **610** can detect and execute scripts and other logic which are embedded in the web resource in order to implement the interactive system **100**

(see FIG. 1A through FIG. 1C). In some of the examples, some of the scripts **615** which are embedded with the web resources **605** can include GPU accelerated logic that is executed directly by the GPU **612**. The main processor **610** and the GPU can combine to render content **611** on a display component **640**. The rendered design interface can include web content from the browser **625**, as well as design interface content and functional elements generated by scripts and other logic embedded with the web resource **605**. By including scripts **615** that are directly executable on the GPU **612**, the logic embedded with the web resource **605** can better execute the interactive system **100**, including the plugin management system **200**, as described with various examples.

CONCLUSION

(151) Although examples are described in detail herein with reference to the accompanying drawings, it is to be understood that the concepts are not limited to those precise examples. Accordingly, it is intended that the scope of the concepts be defined by the following claims and their equivalents. Furthermore, it is contemplated that a particular feature described either individually or as part of an example can be combined with other individually described features, or parts of other examples, even if the other features and examples make no mention of the particular feature. Thus, the absence of describing combinations should not preclude having rights to such combinations.

Claims

1. A network computer system comprising: one or more processors; and a memory to store a set of instructions; wherein the one or more processors access the instructions from the memory to perform operations that include: maintaining a data store that identifies a plurality of plugins; receiving, via a text entry box within a search user interface, a first set of input provided by a user of a computing device; matching the first set of input to a plugin in the data store; determining, based on a set of configurations associated with the plugin, one or more prompts for a set of parametric values needed to execute the plugin; displaying, within the text entry box, a visual representation of a given prompt that specifies at least one characteristic of a parametric value to be provided for the given prompt; displaying, within the search user interface and for the given prompt, a set of selectable options that correspond to the at least one characteristic; determining the set of parametric values based on a second set of input received from the user via the search user interface, wherein the second set of input includes at least a selection of one of the set of selectable options; and triggering execution of the plugin using the set of parametric values in response to detecting a first trigger input provided by the user.
2. The network computer system of claim 1, wherein the first set of input, the second set of input, and the first trigger input are provided by a common input device of the user.
3. The network computer system of claim 1, wherein maintaining the data store includes associating each plugin with plugin execution logic that is specific to the plugin, and wherein the operations further include: upon matching the first set of input to the plugin, executing the plugin execution logic associated with the plugin to display the one or more prompts to the user, the one or more prompts indicating a set of parameter terms associated with the set of parametric values; and receiving the second set of input in response to displaying the one or more prompts.
4. The network computer system of claim 3, wherein the one or more prompts include alphanumeric output that identifies the set of parameter terms.
5. The network computer system of claim 1, wherein the one or more prompts are visually distinguishable from the first set of input within the text entry box.
6. The network computer system of claim 1, wherein the operations include: recording historical information that includes the plugin and the set of parametric values; and displaying, within the search user interface based on the historical information, a preview that includes an indicator of the plugin.

7. The network computer system of claim 6, wherein the indicator includes at least a portion of the set of parametric values.
8. The network computer system of claim 6, wherein the operations include: triggering execution of the plugin after displaying the preview.
9. The network computer system of claim 1, wherein the input characteristic indicates at least one of a format, a structure, or a type of input for the set of parametric values.
10. A method for providing plugins, the method being implemented by one or more processors and comprising: maintaining a data store that identifies a plurality of plugins; receiving, via a text entry box within a search user interface, a first set of input provided by a user of a computing device; matching the first set of input to a plugin in the data store; determining, based on a set of configurations associated with the plugin, one or more prompts for a set of parametric values needed to execute the plugin; displaying, within the text entry box, a visual representation of a given prompt that specifies at least one characteristic of a parametric value to be provided for the given prompt set of parametric values; displaying, within the search user interface and for the given prompt, a set of selectable options that correspond to the at least one characteristic; determining the set of parametric values based on a second set of input received from the user via the search user interface, wherein the second set of input includes at least a selection of one of the set of selectable options; and triggering execution of the plugin using the set of parametric values in response to detecting a first trigger input provided by the user.
11. The method of claim 10, wherein the first set of input, the second set of input, and the first trigger input are provided by a common input device of the user.
12. The method of claim 10, wherein maintaining the data store includes associating each plugin with plugin execution logic that is specific to the plugin, and further comprising: upon matching the first set of input to the plugin, executing the plugin execution logic associated with the plugin to display the one or more prompts to the user, the one or more prompts indicating a set of parameter terms associated with the set of parametric values; and receiving the second set of input in response to displaying the one or more prompts.
13. The method of claim 10, wherein the one or more prompts include alphanumeric output that indicates a type, a format, or a structure of a parametric value in the set of parametric values.
14. The method of claim 10, wherein the one or more prompts are displayed adjacent to an identifier for the plugin within the text entry box.
15. The method of claim 10, further comprising: recording historical information that includes the plugin and the set of parametric values; and displaying, within the search user interface based on the historical information, a preview that includes an indicator of the plugin.
16. The method of claim 15, wherein the indicator includes at least a portion of the set of parametric values.
17. The method of claim 16, further comprising: triggering execution of the plugin after displaying the preview.
18. The method of claim 10, wherein the input characteristic indicates at least one of a format, a structure, or a type of input for the set of parametric values.
19. A non-transitory computer-readable medium that stores instructions, which when executed by one or more processors of a network computer system, cause the network computer system to perform operations that include: maintaining a data store that identifies a plurality of plugins; receiving, via a text entry box within a search user interface, a first set of input provided by a user of a computing device; matching the first set of input to a plugin in the data store; determining, based on a set of configurations associated with the plugin, one or more prompts for a set of parametric values needed to execute the plugin; displaying, within the text entry box, a visual representation of a given prompt that specifies at least one characteristic of a parametric value to be provided for the given prompt; displaying, within the search user interface and for the given prompt, a set of selectable options that correspond to the at least one characteristic; determining the

set of parametric values based on a second set of input received from the user via the search user interface, wherein the second set of input includes at least a selection of one of the set of selectable options; and triggering execution of the plugin using the set of parametric values in response to detecting a first trigger input provided by the user.

20. The non-transitory computer-readable medium of claim 19, wherein maintaining the data store includes associating each plugin with plugin execution logic that is specific to the plugin, and wherein the operations further include: upon matching the first set of input to the plugin, executing the plugin execution logic associated with the plugin to display the one or more prompts to the user, the one or more prompts indicating a set of parameter terms associated with the set of parametric values; and receiving the second set of input in response to displaying the one or more prompts.
